



NEOXYM 500 WSP

MAL no.:

Active Ingredient:

Each gram contains
Neomycin (as sulphate) ----- 500.0mg

Product Description:

White or yellowish-white powder which can be dissolved in animal drinking water

Dosage form:

Water soluble powder

Pharmacodynamics:

Neomycin is an antibiotic from the aminoglycoside family. Aminoglycosides have a broad antibacterial spectrum with good activity against Gram negative species, especially *Escherichia coli* and less activity against Gram positive species. This class of antimicrobials has no effect against anaerobic bacteria.

Neomycin binds to the 30S subunit of the bacterial ribosome which disturbs the reading of the constituent code of the RNA messenger, and finally the synthesis bacterial protein. At high concentrations, it has been shown that aminoglycosides damage the cell wall, conferring bactericidal and bacteriostatic properties.

The resistance mechanisms are complex and differ between aminoglycoside molecules. Four mechanisms of resistance have been identified: changes of the ribosome, reduction of permeability, inactivation by enzymes and substitution of the molecular target. The common mechanism of resistance is the production of aminoglycoside modifying enzymes. These resistance mechanisms can be located in mobile genetics elements increasing the likelihood of spread of aminoglycoside resistance as well as co and cross-resistance. The level of resistance of pathogenic *E. coli* towards neomycin in calves in Europe ranges between 20 and 50 %.

Pharmacokinetics:

Neomycin is poorly absorbed from the gastrointestinal tract. Absorption from the gastrointestinal tract can be significant in neonates. 90% of neomycin is excreted in the faeces after oral administration.

Environmental properties

The active ingredient neomycin sulfate is persistent in the environment.

Indication:

For the treatment and control of colibacillosis (bacterial enteritis) caused by *Escherichia coli* susceptible to neomycin sulfate in cattle (excluding veal calves), swine, sheep and goats.

For the control of mortality associated with *Escherichia coli* organisms susceptible to neomycin sulfate in growing turkeys.

Mode of Administration:

Oral administration via drinking water

Recommended Dosage:

Goats- For the treatment and control of colibacillosis (bacterial enteritis) caused by *E. coli* susceptible to neomycin sulfate. 10 mg of neomycin sulfate per pound of body weight per day is divided doses for a maximum of 14 days. Add to drinking water. Not for use in liquid supplements. Prepare a fresh solution daily. If symptoms persist after using this preparation for 2 or 3 days, consult a veterinarian. Treatment should continue 24 to 48 hours beyond remission of disease symptoms, but not to exceed a total of 14 consecutive days. Discontinue treatment 3 days before slaughter.

Turkeys (growing)- For the control of mortality associated with *E.coli* organisms susceptible to neomycin sulfate in growing turkeys. 10 mg of neomycin sulfate per pound of body weight per day (22mg per kg) for 5 days. Add to drinking water. Not for use in liquid supplements. Prepare a fresh solution daily. If symptoms persist after using this preparation for 2-3 days, consult a veterinarian. Treatment should continue 24 to 48 hours beyond remission of disease symptoms, but not to exceed a total of 5 consecutive days.

Cattle (excluding veal calves)- For the treatment and control of colibacillosis (bacterial enteritis) caused by *E.coli* susceptible to neomycin sulfate. 10mg of neomycin sulfate per pound of body weight per day in divided doses for a maximum of 14 days. Add to drinking water. Not for use in liquid supplements. Prepare a fresh solution daily. If symptoms persist after using this preparation for 2-3 days, consult a veterinarian. Treatment should continue 24 to 48 hours beyond remission of disease symptoms, but not to exceed a total of 14 consecutive days. Discontinue treatment 1 day before slaughter.

Sheep- For the treatment and control of colibacillosis (bacterial enteritis) caused by *E. coli* susceptible to neomycin sulfate. 10 mg of neomycin sulfate per pound of body weight per day is divided doses for a maximum of 14 days. Add to drinking water. Not for use in liquid supplements. Prepare a fresh solution daily. If symptoms persist after using this preparation for 2 or 3 days, consult a veterinarian. Treatment should continue 24 to 48 hours beyond remission of disease symptoms, but not to exceed a total of 14 consecutive days. Discontinue treatment 2 days before slaughter.

Swine- For the treatment and control of colibacillosis (bacterial enteritis) caused by *E. coli* susceptible to neomycin sulfate. 10 mg of neomycin sulfate per pound of body weight per day is divided doses for a maximum of 14days. Add to drinking water. Not for use in liquid supplements. Prepare a fresh solution daily. If symptoms persist after using this preparation for 2 or 3 days, consult a veterinarian. Treatment should continue 24 to 48 hours beyond remission of disease symptoms, but not to exceed a

total of 14 consecutive days. Discontinue treatment 3 days before slaughter.

Contraindications:

Do not use in known cases of hypersensitivity to the active substance, to aminoglycosides or to any of the excipients.

Do not use in cases of intestinal obstruction.

Special Warnings for each Target Species:

Medicated drinking water intake can be affected by the severity of the disease. In case of insufficient intake of water, calves and pigs should be treated parenterally.

Special Precautions for Use in Animals:

Powder for oral solution that is to be dissolved in water and cannot be used as it is. Special care should be taken when considering administration of the product to the newborn calf due to the known higher gastrointestinal absorption of neomycin in neonates. This higher absorption could lead to an increased risk of oto- and nephrotoxicity. The use of the product in neonates should be based on the benefit/risk determination from the attending veterinarian.

Use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria. Official, national and regional antimicrobial policies should be taken into account when the product is used.

Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to neomycin and may decrease the effectiveness of treatment with aminoglycosides due to the potential for cross-resistance.

Special Precautions to be taken by the Person Administering the Veterinary Medicinal Product to Animals:

Wash hands after use. People with known hypersensitivity to aminoglycosides should avoid contact with the veterinary medicinal product.

If you develop symptoms following exposure such as a skin rash, you should seek medical advice and show the doctor this warning. Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention.

Interactions with other Medicaments:

General anaesthetics and muscle relaxing products increase the neuro-blocking effect of aminoglycosides. This may cause paralysis and apnoea. Special care should be taken when using concurrently with diuretics and potentially oto- or nephrotoxic substances.

Usage during Pregnancy and Lactation:

Laboratory studies in animals have not produced any evidence of teratogenic effects of neomycin. The safety of the veterinary medicinal product has not been established during pregnancy, lactation or lay in the target species. Use only according to the benefit/risk assessment by the responsible veterinarian.

Adverse Effects:

None known.

Overdose and Treatment:

Nephrotoxic and/or ototoxic effects may occur in case of accidental overdose.

Withdrawal Periods:

Turkeys: 0 days; Cattle: 1 day; Sheep: 2 days.; Swine and goats: 3 days

A withdrawal period has not been established for this product in pre-ruminating calves. Do not use in calves to be processed for veal. A milk discard period has not been established for this product in lactating dairy cattle. Do not use in female dairy cattle 20 months of age or older. Use of more than one product containing neomycin or failure to follow withdrawal times may result in illegal drug residues.

Storage Condition:

Store in a cool and dry place (< 30°C)

Shelf Life:

36 months

Shelf Life after Opening:

Discard any unused powder immediately after opening

Shelf Life after Dilution/Reconstitution:

24 hours

Packaging Available:

500g/packet, 40 packets/carton, 20kg/carton;

1kg/packet, 20 packets/ carton, 20kg/ carton.

Manufacturer:

N.S. Laboratory Sdn. Bhd.

54, Jalan S2C2, Green Technology Park, Seremban 2, 70300

Seremban, Negeri Sembilan, Malaysia.

Registration Holder:

GNC Agritech Sdn. Bhd.

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Bukit Mertajam Pulau Pinang, Malaysia

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20/07/2025